

FlagOS Online Laboratory

Practical Handbook

Step-by-step guide for setting up and running Qwen3-4B inference on Huawei NPU

1. Login to Online Laboratory Environment

Official Application URL: <https://flagos.io/login>

Official Documentation: <https://docs.flagos.io/projects/onlinelaboratory/en/latest/onlinelaboratory.html>

Query Huawei NPU:

```
source /usr/local/Ascend/ascend-toolkit/set_env.sh

# change the pip mirror
pip config set global.index-url https://pypi.tuna.tsinghua.edu.cn/simple
pip config set global.trusted-host pypi.tuna.tsinghua.edu.cn

npu-smi info
```

```
root@6edf64f2602a:~# npu-smi info
```

NPU		Name	Health	Power(W)	Temp(C)	Hugepages-Usage(page)	
Chip	Phy-ID	Bus-Id	AICore(%)	Memory-Usage(MB)	HBM-Usage(MB)		
7	Ascend910	OK	165.4	41	0	/	0
0	14	0000:81:00.0	0	0	/	0	60030/ 65536
7	Ascend910	OK	-	42	0	/	0
1	15	0000:83:00.0	0	0	/	0	2881 / 65536

NPU	Chip	Process id	Process name	Process memory(MB)
7	0	72637	VLLMEngineCor	56880
7	0	72090	vllm	70

2. Install Qwen3-4B Model

```
pip install modelscope
# if there is a setup problem, downgrade the setup tools
# pip install "setuptools<81"
```

Download the model weights of Qwen3-4B officially open-sourced by Qwen via modelscope:

```
# takes 5 min
modelscope download --model Qwen/Qwen3-4B --local_dir ~/Qwen3-4B
```

3. Install FlagGems, FlagTree, vllm-plugin-FL

Install FlagGems:

```
pip install flag_gems
```

Verify installation successful:

```
pip show flag_gems
```

```
Name: flag_gems
Version: 4.2.1rc0
Summary: FlagGems is a function library written in Triton.
Home-page: https://github.com/FlagOpen/FlagGems
Author:
Author-email: Zhixin Li <strongspoon@outlook.com>, Tongxin Bai <waffle.bai@gmail.com>, Yuming Huang <jokmingwong@gmail.com>, Feiyu Chen <iclementine@outlook.com>
License: Apache Software License
Location: /usr/local/python3.11.13/lib/python3.11/site-packages
Editable project location: /root/FlagGems
Requires: packaging, pybind11, PyYAML, sqlalchemy, torch
Required-by:
```

Install FlagTree:

```
RES="--index-url=https://resource.flagos.net/repository/flagos-pypi-hosted/simple --
trusted-host=https://resource.flagos.net"
python3 -m pip install flagtree==0.4.0+ascend3.2 $RES
```

Verify installation successful:

```
pip show flagtree
```

```
Summary: A unified compiler supporting multiple AI chip backends for custom Deep Learning operations, which is forked from triton-lang/triton.
Home-page: https://github.com/triton-lang/triton/
Author: FlagOS
Author-email: contact@flagos.io
License:
Location: /usr/local/python3.11.13/lib/python3.11/site-packages
Requires:
Required-by:
```

Install vllm-plugin-FL:

```
git clone --branch "v0.1.1+vllm0.13.0" --depth 1 https://github.com/flagos-ai/vllm-
plugin-FL.git
cd vllm-plugin-FL
pip install -r requirements/ascend.txt
pip install --upgrade setuptools
pip install --no-build-isolation -e .
```

Verify software version:

```
cd vllm-plugin-FL
git log
```

```
root@6edf64f2602a:~/vllm-plugin-FL# git log
commit ba008211c1c9646e19290e832a7f7775f7d2944f (HEAD -> main, origin/main
, origin/HEAD)
Author: xin2an <yuxinan@perfxlab.com>
Date: Sat Feb 28 22:02:09 2026 +0800

    The Ascend platform adds a new Flagos blacklist operator. (#63)

### PR Category
<!-- One of [Core | Vendor | OP | Tools | Others] -->
```

4. Configure Inference Service

Start service: Need to modify the model path, e.g., /root/Qwen3-4B

```
export VLLM_PLUGINS=fl
export TRITON_ALL_BLOCKS_PARALLEL=1
vllm serve --model /root/Qwen3-4B --served-model-name qwen --enforce-eager --port 30000
```

Note: When starting the inference service, FlagGems operators will be automatically started. You can query the enabled FlagGems operators with the following command:

```
cat /tmp/flaggems_enable_oplist.txt
```

5. Call Inference Service Within Laboratory

```
pip install openai
touch test.py
# modify python file
python test.py
```

```

from openai import OpenAI

client = OpenAI(
    base_url="http://localhost:30000/v1",
    api_key="any"
)

response = client.chat.completions.create(
    model="qwen",
    messages=[{"role": "user", "content": "Introduce yourself"}],
    temperature=0.7,
    max_tokens=512
)

print(response.choices[0].message.content)

```

6. Call Inference Service Outside Laboratory

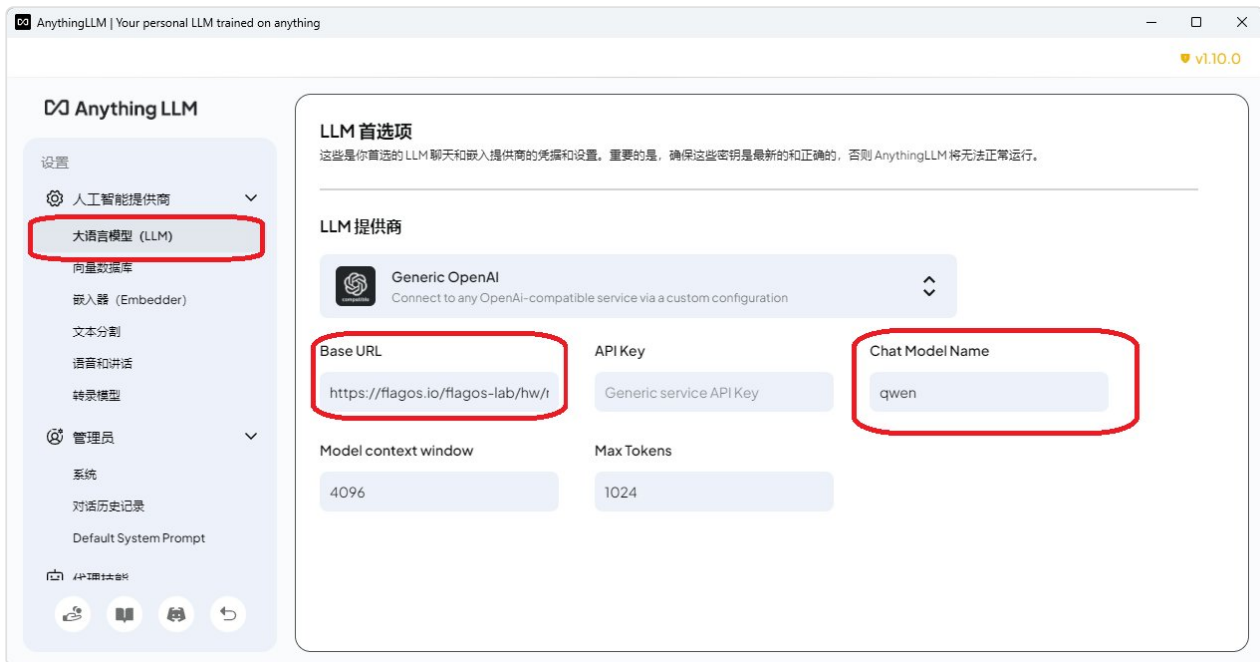
If you configured port 30000 when starting the service, you can connect to the laboratory server from outside the laboratory using a client to create a bot for conversation. Steps are as follows:

Step 1: Download AnythingLLM.

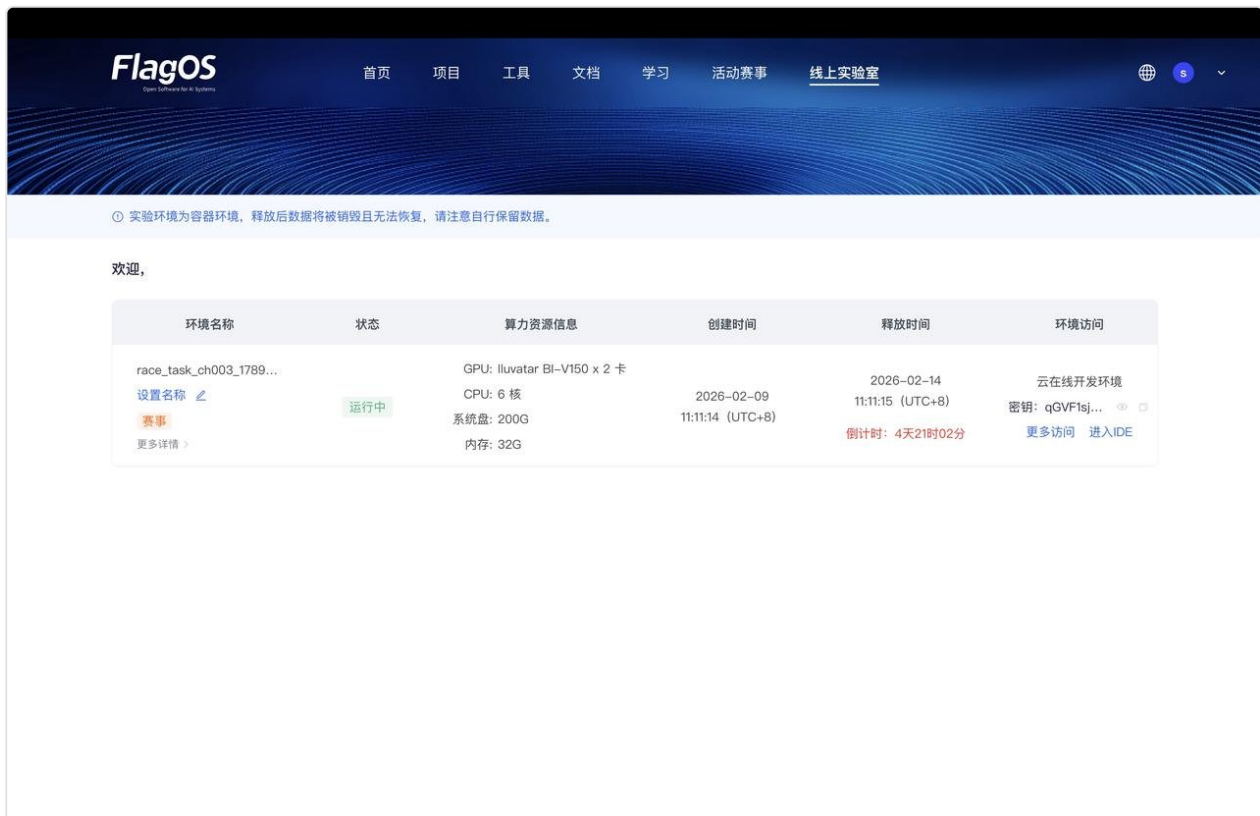
Step 2: Click the wrench button at the bottom-left of the interface:



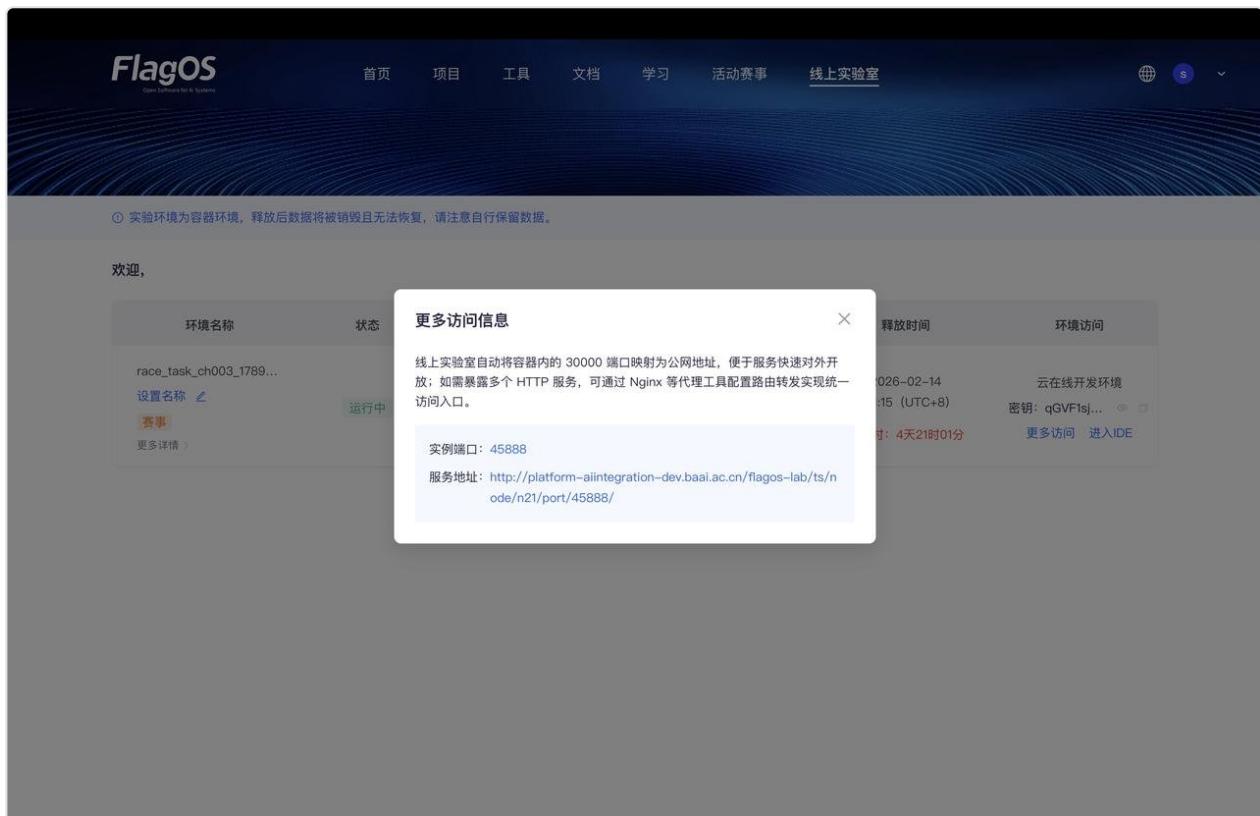
Step 3: Select **Large Language Model (LLM)** and configure required items: Base URL and Chat Model Name. API Key is not required. Others are optional.



Base URL: Fill in the Service Address from when you applied for the online laboratory. Return to the online laboratory and click **More Access**:



In the pop-up dialog, copy the Service Address, and add **v1** after the service address to form a complete URL. Paste it into AnythingLLM's Base URL.



Chat Model Name: Fill in the name **qwen** from the command used when starting the service:

```
vllm serve --model /root/Qwen3-4B --served-model-name qwen --enforce-eager --port 30000
```

Click the wrench to return to My Workspace. In My Workspace, click **New Thread**, name it "Bot", and have a conversation in the message window.

